

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (EE) Sem -IV Examination, 2010-11

EE205 Electrical Machines – I

Date: 10/05/2011, Tuesday

Time: 10:00 am to 01:00 pm

Maximum Marks:70

Instructions:

- [1] All the questions are compulsory to attempt.
- [2] Figures to the right indicate full marks.
- [3] Assume suitable data wherever necessary.

SECTION- I

- Q1(A) Explain the working principle and construction of a 1-phase transformer. 04
- (B) The following test results were obtained for a 20 kVA, 50 Hz, 2400/240 V distribution transformer. 06

OC Test: 240V, 1.066 A, 126.6 W

SC Test: 57.5 V, 8.34 A, 284 W

- (i) When the transformer is operated as a step down transformer with the output voltage equal to 240 V, supplying a load at unity power factor, determine the maximum efficiency and the unity power factor load at which it occurs.
- (ii) Determine the power factor of the rated load, supplied at 240 V such that the terminal voltage observed on reducing the load to zero is still 240 V.

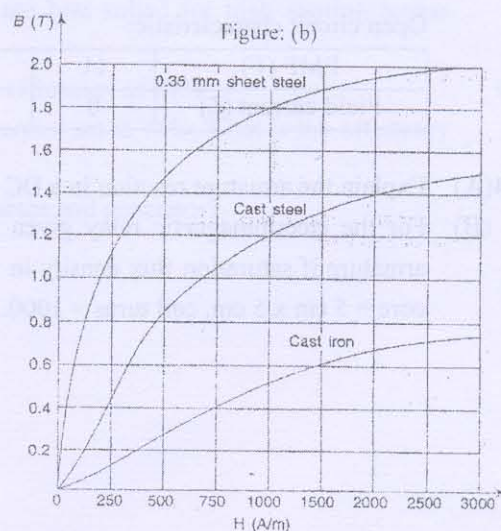
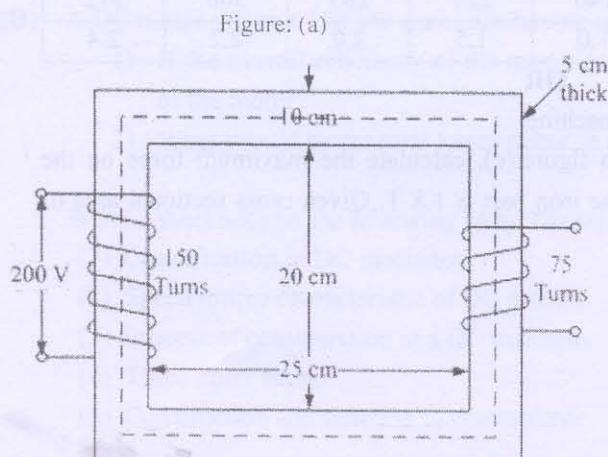
(C) Attempt any *two*. 04

- 1) Justify the statement: 'Primary and secondary currents are inversely proportional to each other.'
- 2) Why is an SC test performed on HV side and at a reduced voltage?
- 3) Elaborate: 'The core flux in a transformer is independent of load current.'

OR

- Q1(A) Derive an equivalent circuit of a 1-phase transformer. Also, draw a phasor diagram for the same. 06

- (B) The core of a transformer shown in figure:(a) is made of silicon sheet steel (B-H curve is shown in figure: (b)). Calculate the no load current with primary excited at 200 V, 50 Hz. Assume core loss to be 230 kW/m³ (of core volume). What is the no load power factor and the magnitude of no load power drawn from the source. 08



Q2(A) Name the four phasor groups pertaining to 3-phase transformer. Draw the connection schemes and phasor diagrams for each of these four groups. 05

(B) Two single-phase furnaces supplied at 250 V from a 6.6 kV, 3-phase system through a pair of Scott connected transformers. If the load on the main transformer is 85 kW at 0.9 power factor lagging and that on teaser transformer is 69 kW at 0.8 power factor lagging, find the values of the line currents on the 3-phase side. Also, draw the phasor diagram. 06

OR

Q2(A) Prove that in open delta connection, 05
Power supplied by each transformer = $86.66\% V_L I_L$.
Where V_L and I_L having the usual notations.

(B) Attempt any *three*. 06

1) Justify the statement: 'An auto transformer cannot be used as distribution transformer.'

2) Why a transformer is rated in terms of kVA or MVA?

3) Elaborate: 'The transformer tapplings are usually provided on HV side.'

4) Why the core loss of a distribution transformer is less as compared to a power transformer of a same rating?

Q3 Write a short note on the following [Any *Two*] 10

(1) Auto transformer

(2) Tertiary winding

(3) Conditions for the parallel operation of 1-phase transformers

(4) Factors affecting the choice of 3-phase transformer connections

SECTION – II

Q4(A) For a singly excited magnetic system, establish relationship between magnetic field energy and co energy. 05

(B) Calculate the efficiency of a 10 kW, 250 V, 1000 rpm self-excited DC shunt generator from the following data: 05

Armature resistance: 0.35Ω

Voltage droop at brushes: 2 V

Windage and friction losses: 150 W

Iron loss at 250 V = 180 W

Open circuit characteristic:

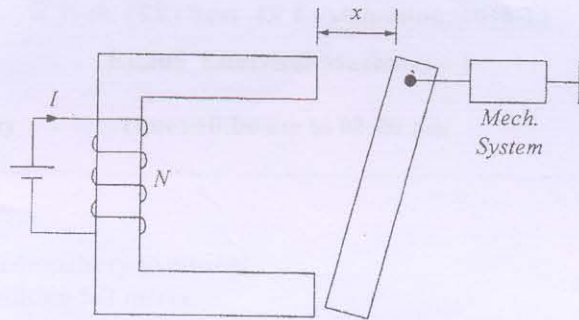
EMF (V)	11	140	227	285	300	312
Field current (I_f)	0	1.0	1.5	2.0	2.2	2.4

OR

Q4(A) Explain the armature reaction in a DC machine. 05

(B) For the electromagnetic relay given in figure(c), calculate the maximum force on the armature if saturation flux density in the iron part is 1.8 T. Given cross sectional area of core = 5 cm x 5 cm, coil turns = 1000. 05

Figure: (c)



Q5(A) Give the reasons for failure of voltage building up process in case of self-excited DC generator. What are the remedial measures to overcome the failure of voltage building up process? 05

(B) A 20 HP, 230 V, 1150 rpm DC shunt motor has four poles and 882 lap connected conductors. The armature circuit resistance is 0.188Ω . At rated speed and rated output the armature current is 73 A and field current is 1.6 A. 05

Calculate:

- 1) Electromagnetic torque
- 2) Flux per pole
- 3) Rotational loss
- 4) Efficiency
- 5) Shaft load

OR

Q5(A) Attempt any *three*. 06

- 1) Elaborate the statement: 'Energy and co energy of a linear magnetic field has equal numerical values.'
- 2) Explain why the external characteristic of a DC shunt generator is more drooping than that of a separately excited DC generator.
- 3) Define leakage flux. Does it take part in electromechanical energy conversion process via the magnetic field?
- 4) Explain why differentially compound motor is not used in practice.
- 5) Justify the statement: 'DC series motor ate best suited for high starting torque load.'

(B) A DC motor drives a 100 kW generator having an efficiency of 87% 04

- 1) If the overall efficiency of the motor-generator set is 74%, what is the efficiency of the motor?
- 2) What should be the total losses occur in motor and generator?

Q6 Write a short note on the following [Any *Three*] 15

- (1) Classification of DC machines
- (2) Speed torque characteristic of DC motors
- (3) Process of commutation in a DC machine.
- (4) Three point starter
- (5) Construction and function of commutator